

G. Lejeune Dirichlet's Werke. II.

XVIII.

On the equation $t^2 + u^2 + v^2 + w^2 = 4m$

Translated from the French

Liouville, Journal de Mathématiques pures et appliquées, Série II, Tome I, 1856, p. 210–214.

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[Extract from a letter of M. Lejeune Dirichlet to M. Liouville.]

Permit me, dear friend, to return for a moment to the conversation we had recently about Jacobi's beautiful theorem on the number of decompositions of an integer into four squares, a theorem which that illustrious geometer first drew from his elliptic series and for which he later gave an arithmetical proof. Having better recalled my memories, I shall present to you, with more development and especially in somewhat better order, what the other day I could only indicate. As I was saying to you, I have always experienced a certain difficulty, which I believe has also occurred to other persons, when I wanted to reproduce this beautiful proof without having the author's text before my eyes and without recurring to the operations on series of which Jacobi's proof, as he himself warns, is only the translation. This circumstance led me long ago to seek a new proof of the theorem in question on other principles; but I shall not speak of it here, since that proof will find its natural place in a work whose writing has occupied me for some time. For the moment, my only object is to simplify Jacobi's proof and especially to make it easier to retain, by bringing into full light the arithmetical, or rather algebraic, fact which forms its principal foundation.

To avoid useless repetitions, I shall note that all the integers which we denote by Latin letters are positive and odd, except those denoted by x and x' , which are not subject to this restriction.

As for the lemma which serves as Jacobi's point of departure and which defines the number of solutions of the equation

$$t^2 + u^2 = 2p,$$

p being supposed given, I shall not reproduce here the very simple demonstration that Jacobi gave of it, all the more since this lemma falls within a general proposition of which I have had occasion to speak elsewhere. It is known that one must perform all decompositions

$$p = ad$$

of p into two factors, and that the number in question is equal to the excess of the number of cases in which the first factor a is of the form $4\nu + 1$ over the number of cases in which a is of the form $4\nu + 3$. Thus, agreeing to make correspond to each decomposition a unit δ , positive or negative according to these two cases, the number in question will be expressed by the sum

$$\sum \delta.$$

This being established, let us try to determine the number of solutions of the equation

$$t^2 + u^2 + v^2 + w^2 = 4m, \tag{1}$$

in which m denotes a given odd integer. To obtain all these solutions, one must set, in all possible ways,

$$4m = 2p + 2q \quad \text{or} \quad 2m = p + q,$$

and then solve, for each pair p, q , the two equations

$$t^2 + u^2 = 2p, \quad v^2 + w^2 = 2q$$

in the most general manner.

If, first considering a fixed pair p, q , we make correspond to each decomposition

$$q = bc$$

a unit $\varepsilon = \pm 1$, which depends on b in the same manner as δ depends on a , the number of solutions of equation (1) corresponding to this pair p, q will be

$$\sum \delta \cdot \sum \varepsilon,$$

or also

$$\sum \eta,$$

the terms of this sum corresponding to the different combinations that can be formed with the decompositions

$$p = ad, \quad q = bc,$$

and η being equal to the positive or negative unit according as the residues of a and b , when divided by 4, are or are not equal; equivalently, according as $a - b$ is or is not divisible by 4. With this understood, it is manifest that the number of all solutions of our equation (1) will again be expressed by the sum

$$\sum \eta,$$

whose terms will always depend in the same manner on a and b , but will now correspond to all combinations which a and b present in the complete solution of the equation

$$2m = ad + bc. \tag{2}$$

Before going further, it is useful to make a very simple remark which will be useful to us later: among the two even integers

$$a - b \quad \text{and} \quad c + d$$

there is always one, and only one, which is divisible by 4. Indeed, if it were otherwise, one would have

$$\text{either } a \equiv b, \ c \equiv -d, \quad \text{or } a \equiv -b, \ c \equiv d \pmod{4},$$

and it would follow that $ad + bc \equiv 0 \pmod{4}$. Thus η may be defined otherwise by saying that η is equal to -1 or $+1$, according as $c + d$ is or is not divisible by 4.

Let us now distribute the solutions of equation (2) into two classes, putting into the first those for which $a = b$, and into the second all the others. Since the latter are associated two by two and result one from the other by exchanging simultaneously a with b , and d with c , which does not change the value of η , one sees that the number H of solutions of equation (1) can be put in the form

$$H = \sum \eta' + 2 \sum \eta'',$$

where the terms η' of the first sum, which moreover are all equal to the positive unit, correspond to the solutions of equation (2) for which $a = b$, and the terms η'' of the second correspond to the solutions of the same equation that satisfy the condition $a > b$.

This being done, we shall first consider the solutions of (2) for which $a > b$. Put

$$\begin{aligned} a' &= c(x+1) + d(x+2), & c' &= a(x+1) - b(x+2), \\ b' &= cx + d(x+1), & d' &= -ax + b(x+1). \end{aligned}$$

Then identically

$$a'd' + b'c' = ad + bc.$$

It follows that every solution of equation (2) supplies infinitely many others by means of the indeterminate integer x ; but it remains to see to what restriction this integer must be subject in order that a', b', c', d' be odd and positive, and that in addition $a' > b'$ as in the primitive solution. Now each of the pairs

$$x, x+1 \quad \text{and} \quad x+1, x+2$$

is composed of two integers, one even and the other odd, so the first condition is always fulfilled. As for the second, since

$$c' = (a-b)(x+1) - b, \quad d' = b - (a-b)x,$$

one sees that it completely determines the integer x : $(a-b)x$ must be the multiple of the positive even number $a-b$ immediately below the odd number b . One also sees, since x is not negative, that a' and b' will be positive and that, moreover, $a' > b'$.

Having assured ourselves that the expressions given above always furnish a unique solution subject to the same conditions as the primitive solution, let us see to what solution we shall be led if we take the solution a', b', c', d' in turn as point of departure. For this, in the expressions

$$\begin{aligned} c'(x'+1) + d'(x'+2), & \quad a'(x'+1) - b'(x'+2), \\ c'x' + d'(x'+1), & \quad -a'x' + b'(x'+1), \end{aligned}$$

we must give to the indeterminate x' the value, wholly determined by what precedes, which makes these expressions positive. On the other hand, the equations set above give by their solution

$$\begin{aligned} a &= c'(x+1) + d'(x+2), & c &= a'(x+1) - b'(x+2), \\ b &= c'x + d'(x+1), & d &= -a'x + b'(x+1), \end{aligned}$$

whose first members are positive. Thus x' coincides with x , and we are brought back to the primitive solution a, b, c, d .

If now we observe that to our two solutions so linked with one another, and which are mutually deduced one from the other by a uniform procedure, there always correspond opposite values of η'' , as follows from the equation

$$a - b = c' + d'$$

and from the remark made above, we conclude that $\sum \eta''$ reduces to zero, and we simply have

$$H = \sum \eta'.$$

To evaluate this sum, all of whose terms η' are equal to the positive unit, everything reduces to finding the number of solutions of the equation

$$a(d+c) = 2m.$$

Now if a is a fixed divisor of m , $m/a = e$ will also be such a divisor, and one will have to solve the equation

$$d+c = 2e,$$

whose number of solutions is e . Hence, since a , and consequently also e , must successively be made equal to all divisors of m , the number H that was to be determined coincides with the sum of the divisors of m .